HAOKUN CHEN

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EDUCATION

Ph.D. Computer Science, University of Munich (LMU), Germany *11.2021 - Present*
Interests: **Multimodal Learning**: Post-training of LLMs, Vision-Language Foundation Models.
Trustworthy AI: Federated Learning, Machine Unlearning, Adversarial Attacks.

M.Sc. Computer Science, Technical University of Munich, Germany *10.2018 - 10.2021*

B.Sc. Mechatronic Engineering, Tongji University, China *10.2014 - 10.2018*

WORK EXPERIENCE

Machine Learning Researcher, Siemens AI Lab (Munich, Germany) *01.2021 - Present*

- Contributed to building cloud-based Machine Learning infrastructure for RAG-based industrial foundation models finetuning using Microsoft Azure ML.
- Developed a hyperparameter tuning pipeline for (Multimodal) LLM post-training across multi-node, multi-process distributed systems; manuscript accepted at *AAAI 2025* (Oral, First Author).
- Proposed a PEFT-based post-training algorithm for Vision-Language Models (VLMs) using privacy-preserving federated learning across heterogeneous distributed datasets; manuscript accepted at *AAAI 2024* (First Author).
- Designed algorithms to address data heterogeneity, data scarcity, and system scalability challenges in federated learning; manuscripts accepted at *ICCV 2023* and *CVPR 2025* (Both First Author).
- Conducted research on adversarial robustness and red-teaming strategies for multimodal LLMs.
- Contributed to customer-focused data science and ML projects across various business units.
- Provided technical leadership and mentorship to master's students and junior PhD researchers.

Research Scientist Intern, Intel Labs (US, Remote) *08.2024 - 11.2024*

- Implemented a novel algorithm and an evaluation framework to systematically audit existing machine unlearning methods for LLMs.
- Contributed to LLM adversarial robustness toolkit (LLMart): *https://github.com/IntelLabs/LLMart* to facilitate the evaluation of adversarial attack algorithms for LLMs.

Research Intern, BMW Autonomous Driving Campus (Munich, Germany) *03.2020 - 11.2020*

- Contributed to the development of validation frameworks for computer vision functions in autonomous driving, using C++.
- Conducted research on data augmentation techniques for traffic sign detection.

Software Engineer Intern, BMW Research Center (Beijing, China) *08.2019 - 12.2019*

- Contributed to the development of validation frameworks for in-vehicle entertainment systems.
- Visualization and processing of validation results using Visual Basic.

PUBLICATIONS

Haokun Chen, et al., FedBiP: Heterogeneous One-Shot Federated Learning with Personalized Latent Diffusion Models, *CVPR 2025*.

Haokun Chen, et al., FedPop: Federated Population-based Hyperparameter Tuning, *AAAI 2025*, **Oral**.

Haokun Chen, et al., FedDAT: An Approach for Foundation Model Finetuning in Multi-Modal Heterogeneous Federated Learning, *AAAI 2024*.

Haokun Chen, et al., FRAug: Tackling Federated Learning with Non-IID Features via Representation Augmentation, *ICCV 2023*.

Haokun Chen, et al., Soft Token Attacks Cannot Reliably Audit Unlearning in Large Language Models, *Under review*, 2025.

Haokun Chen, et al., AUVIC: Adversarial Unlearning of Visual Concepts for Multi-Modal Large Language Models, *Under review*, 2025.

Haokun Chen, et al., Does Gradient Ascent Truly Remove Model Knowledge? A Framework for Auditing Unlearning in LLMs, *Under review*, 2025.

Yao Zhang, Haokun Chen, et al., Cl-crossvqa: A continual learning benchmark for cross-domain visual question answering, *WACV 2025*.

Jinhe Bi, Yujun Wang, Haokun Chen, et al., LLaVA Steering: Visual Instruction Tuning with 500x Fewer Parameters through Modality Linear Representation-Steering, *Under review*, 2025.

Yao Zhang, Hewei Gao, Haokun Chen, et al., FedNano: Toward Lightweight Federated Tuning for Pretrained Multimodal Large Language Models, *Under review*, 2025.

Yao Zhang, Chenyang Lin, Haokun Chen, et al., SwarmAgentic: Towards Fully Automated Agentic System Generation via Swarm Intelligence, *Under review*, 2025.

Ahmed Frikha*, **Haokun Chen***, **et al.**, Towards Data-Free Domain Generalization, *ACML 2022*.

SKILLS

Programming Languages: Python, C++, Unix, Git.

Frameworks: PyTorch, Azure ML, Accelerate, DeepSpeed, Ray-Tune, TernsorRT-LLM, Docker.

Languages: English, Mandarin, German (C1).

REVIEW

Conferences: *Computer Vision*: CVPR 2025, ECCV 2024, ICCV 2025.
Machine Learning: NeurIPS 2024, ICLR 2025, ICML 2025.
Aritificial Intelligence: AAAI 2024/2025, AISTATS 2025, IJCNN 2025.

Journals: TPAMI, TKDE, TKDD, TCNN